Week 5: What Affects the Accuracy of the Simple Transformer?

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1 Setup

The task is per-user rating prediction (five classes, rating -1) from a movie's 19 genre flags on MovieLens 100K. We compare logistic regression (LR, 100 parameters) with the simple transformer (one attention layer, embedding dimension $D = 16$, 2,901 parameters). This week's changes:

- **Training time:** 300 epochs with early stopping disabled (previously the run stopped by epoch 28, so 300 never ran).
- **Data quality:** only users with more than 100 ratings (361 of 943 users); the old code silently defaulted to user 405.
- **Consistent timing:** every user's ratings are split chronologically 70/15/15, so training is strictly earlier than validation, which is strictly earlier than test; this is enforced by a test over all 361 users.
- **Metrics:** mean, standard deviation, 5th and 95th percentiles across users, and a 99% bootstrap confidence interval of the mean (wider than 95%).

Test accuracy is reported at the epoch with the lowest validation loss (the rule was fixed in advance) and at epoch 300. Because users differ widely in how often they give their most common rating, each user's accuracy is also reported as *lift* over their majority-class baseline.

2 Spread of accuracy across users

	Mean	Std	5th pct	95th pct	99% CI of mean
Majority baseline	32.8	17.6	8.2	63.9	[30.6, 35.2]
LR	33.7	14.8	11.5	60.5	[31.7, 35.7]
Transformer	35.4	15.2	13.0	61.3	[33.4, 37.4]

Table 1: Test accuracy (%) at the best-validation epoch, 361 users with more than 100 ratings.

Accuracy varies enormously between users (5th to 95th percentile spans about 48 points), and most of that variation is the baseline itself. Relative to each user's majority baseline, the transformer gains +2.5 points (99% CI $[1.0, 4.0]$) and LR gains +0.9 ($[-1.0, 2.7]$, not distinguishable from zero). The transformer beats LR by +1.7 points on paired users ($[0.3, 3.1]$; better for 45% of users, worse for 33%). Genre flags alone are weak features, so both models are only slightly above "always predict the most common rating."

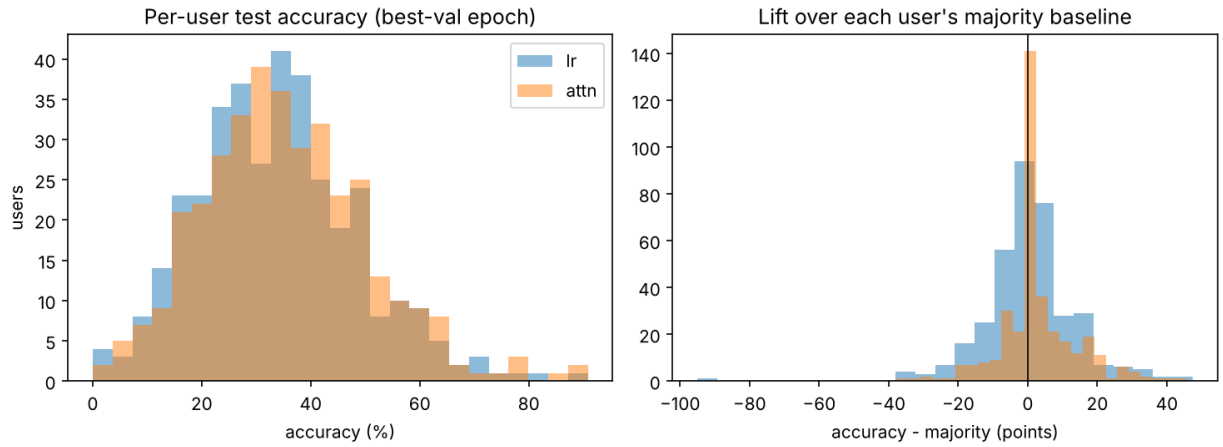


Figure 1: Per-user test accuracy (left) and lift over the majority baseline (right).

3 Training time: plateau and optimum

	Best epoch (median / mean)	Optimal epoch	Plateau	Test acc best-val	Test acc epoch 300
LR	71 / 95	85	from epoch 58	33.7	33.6
Transformer	6 / 11	15	none, then declines	35.4	31.1

Table 2: Optimal epoch is the maximum of the cross-user mean validation accuracy; plateau is the first epoch from which that mean stays within 0.5 points of its maximum for 10 epochs.

More training does *not* help the transformer: its mean validation accuracy is highest in the first few epochs and then declines, and 300-epoch test accuracy is 4.3 points below the best-validation checkpoint (paired difference vs. LR flips from +1.7 to −2.5). LR is the opposite: it climbs from 23% to a plateau by about epoch 58 and stays flat. The best training time is therefore short for the transformer (about 15 epochs, chosen per user by validation loss) and about 60 to 85 epochs for LR. The extra capacity of the transformer overfits each user's small training set.

4 Effect of removing users with 100 or fewer ratings

	Users	Median test size	Accuracy std (LR / Transf.)	Transf. lift
More than 100 ratings	361	27	14.8 / 15.2	+2.5
100 or fewer ratings	516	6.5	24.1 / 25.4	+0.2

Table 3: Kept versus dropped users (582 users have 100 or fewer ratings; 533 of them have at least 10 distinct timestamps, and 17 of those still could not be split chronologically, leaving 516).

The filter mainly removes noise: dropped users have a median of 6.5 test ratings (as few as 1), so their accuracy swings widely (standard deviation about 25 versus 15 points). The difference in lift between the two groups is +2.4 points for the transformer ($[-0.4, +5.1]$) and +2.2 for LR

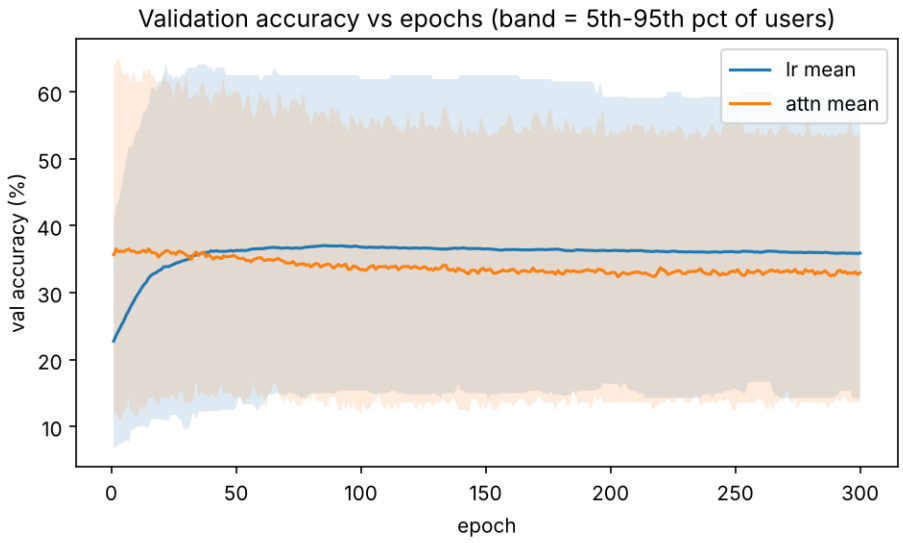


Figure 2: Mean validation accuracy versus epoch; the band is the 5th to 95th percentile across users.

([−1.1, +5.3]); neither is significant at 99%, so we cannot claim the filter raises accuracy, only that it gives cleaner and less variable estimates.

5 Movie synopsis context

Method. Each movie's Wikipedia plot (Wikipedia Movie Plots dataset, matched to MovieLens by title and year; 1,236 of 1,682 movies, 73.5%, covering about 90% of ratings) is turned into a 16-dimensional vector (TF–IDF, then SVD, using plot text only and no ratings). The transformer receives it as one extra context token alongside the 19 genre tokens ($D = 16$, 3,253 parameters); LR receives it as 16 extra features. Movies without a plot get a zero vector. All models use the same 361 users, splits, seeds, and 300 epochs. **Result.** Synopsis context does not measurably improve

	Mean acc best-val	Lift over majority	Paired diff vs. genre-only	Optimal epoch
LR, genres only	33.7	+0.9	—	85
LR + synopsis	34.8	+2.0	+1.1 [−0.2, +2.4]	51
Transformer, genres only	35.4	+2.5	—	15
Transformer + synopsis	34.6	+1.7	−0.8 [−1.7, +0.1]	7

Table 4: Effect of synopsis context (361 users; paired differences in test accuracy points with 99% CI).

accuracy. For the transformer the difference is slightly negative (−0.8 points, 99% CI includes zero); for LR it is slightly positive at the best-validation epoch (+1.1, CI includes zero) and significantly negative at epoch 300 (−1.1). Per-user accuracy is better for 24% of users and worse for 32% with the transformer. The extra token also shortens the best training time (optimal epoch 15 to 7), consistent with faster overfitting from more input dimensions on tiny per-user training sets. A plausible reason is that a 16-dimensional text summary adds many features to learn from a few

hundred training ratings, and plot topics overlap heavily with genre. This is a negative result for this specific encoding, not proof that text context cannot help: richer text encoders or pooling information across users are untested.

6 Why did the paper not use deep neural networks?

Based on the paper's own framing, there are three reasons, none of them that deep models predict better. *First*, the goal is not only prediction but *optimizing* over the model's output in real time (choosing the best set of items among catalogs of up to hundreds of millions, in milliseconds). For general transformers this optimization is NP-hard, so the authors restrict to a simple single-layer transformer for which they prove a near-optimal, sub-linear-time algorithm. *Second*, this restriction is argued to cost little accuracy: the paper reports the one-layer model about 2.5% below the average general (multi-layer) transformer, and on the Spotify data 0.702 accuracy versus 0.704 for two layers and 0.726 for three (Sec. 5.1; figures as summarized from the paper's HTML version, worth a check against its Table 1). *Third*, the analysis assumes a small embedding dimension, which holds in practice. Our own results are consistent with the accuracy point: on small per-user data, extra capacity overfits, so a larger network would not help here either.

7 Limitations and next steps

Per-user test sets are small (median 27 ratings), accuracy is dominated by each user's majority class, and genre flags carry little signal. Next: pool information across users (the paper's setting), try pretrained text embeddings for the synopsis, and report accuracy as lift over the majority baseline throughout.